

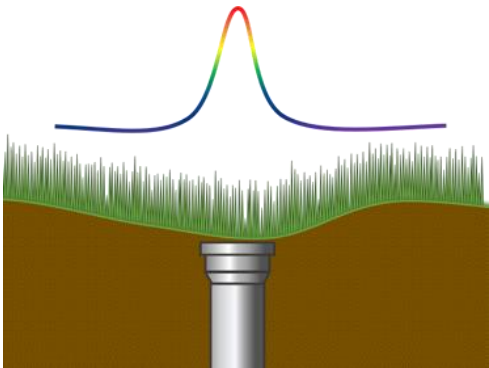
Identifying abandoned well sites using database records and aeromagnetic surveys

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ABSTRACT

Oil and natural gas are primary sources of energy in the United States. Improved drilling and extracting techniques have led to a renewed interest in historic oil and gas fields, but limited records of legacy wells make new drilling efforts more difficult, as abandoned wells may provide conduits for liquids and gases to migrate into groundwater reservoirs or the atmosphere. Well finding using aeromagnetic surveys pinpoints the location of steel-cased wells, detecting both active and abandoned wells, including buried casings lacking aboveground markers. Here, we present six aeromagnetic surveys conducted in Pennsylvania and Wyoming as case studies, comparing the magnetic points to locations known in databases. In all study sites, more magnetic points were detected than recorded in databases. Based on differences between theoretical database well counts and the actual number of wells detected in surveys, we estimated the total number of wells in Pennsylvania to be 395,000 – 466,000 and 181,000 – 182,000 in Wyoming. Extrapolating to the national level, we estimate the average number of wells in the continental United States is 6.04 ± 19.97 million wells with 1.16 ± 3.84 million of those designated as abandoned wells, within the range of previous abandoned well count estimations. Although aeromagnetic surveys are limited to detecting steel-cased wells and do not differentiate sites based on well status, this study nevertheless demonstrates the utility of aeromagnetic surveys in well finding efforts across the country and shows limitations in database records of oil and natural gas wells.



INTRODUCTION

The United States relies on oil and natural gas (O&G) for 66% of its total annual energy consumption, providing nearly 64.5 quadrillion Btu per annum². O&G have been extracted for commercial purposes since 1859, when the first commercially successful oil well was drilled in Titusville, PA. Since then, many thousands of oil wells have been abandoned due to many reasons, including changes in demand, production decline, and business migration to more productive oil fields. Abandoned wells, or wells that remain unplugged or improperly plugged, present not only a safety and environmental hazard but also a problem for current and future land use as new wells are being drilled for hydraulic fracturing to increase oil production from previously active oil fields.

Abandoned oil and gas wells plague current drilling and extraction activities by acting as potential conduits for contamination of groundwater and surface water³⁻⁴. Due to these risks, as well as possible asphyxiation hazards in dwellings near drilling sites, all abandoned wells within 1000 m of a planned unconventional well in Pennsylvania must be properly documented, monitored, and if deemed necessary, sealed⁵. However, online databases and historical records do not contain the accurate or complete record of

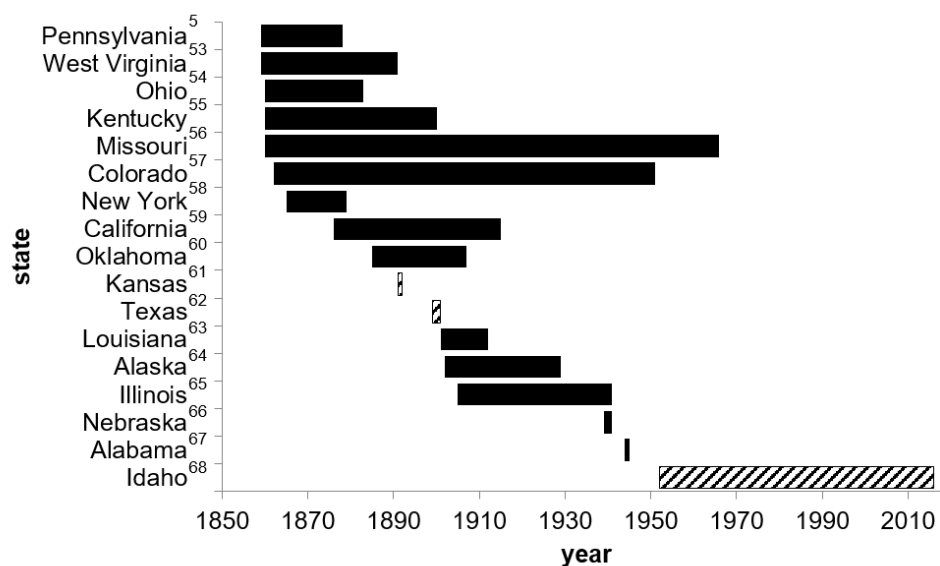


Figure 1. Years between first commercial oil well¹ and first plugging regulation in some states with commercial oil activity. Solid bar, oil activity preceded regulations. Hatched bar, regulations set in place before first successful commercial drilling.

all oil and gas wells drilled in the United States⁶. Often, regulation of drilling activity followed decades of undocumented activity (Figure 1), resulting in incomplete well records. Prior to modern global positioning service (GPS), recorded well locations were notoriously undetailed, further compromising the accuracy of well databases. In addition, locating abandoned wells in the field presents a challenge especially in difficult terrain or densely vegetated regions.

Magnetic surveys provide field evidence for well locations where documentation is lacking. Magnetometers are used to map magnetic fields and to find anomalies or perturbations to the Earth's magnetic field. Ferromagnetic objects on the surface or shallowly buried beneath the surface, such as fenceposts, culverts, pipelines, and well casings, cause a perturbation to the magnetic field. Historically, oil and gas wells have been constructed with steel, although the earliest wells were often made with wood⁷. Frischknecht and colleagues were the first to demonstrate that vertical steel well casings produced a single vertical dipole, appearing as a monopole, in a magnetic intensity map⁸. Thus, wells show up as a monopole in magnetic maps, making them easily differentiable from other ferromagnetic objects that may be present in an oilfield. By analyzing aerial photographs, historic maps, and reports, false positives from other ferromagnetic sites can be eliminated to provide a database of coordinates that likely correspond to a well target, which can then be validated with a field survey⁹.

The power of magnetometry to identify steel-cased wells lies in its ability to detect wells with no visible aboveground infrastructure: even though the casing may be cut off up to 10 ft (3 m) belowground to comply with modern plugging requirements or so that the casing may be used elsewhere^{6, 10-11}, the well can still be detected. The magnetic intensity of well casings is strong enough that even wells buried below 30 m of displaced mine spoil have been detected⁹. It has even been suggested that the intensity of the magnetic anomaly may be used to determine the depth at which the well casing has been cut¹². Well location efficacy based on aerial magnetic (hereafter aeromagnetic) surveys is quite high. Frischknecht estimated that over 95% of wells located across five survey sites in Oklahoma and Colorado were detected during hereafter aeromagnetic surveys¹³. Based on estimates from state databases, historical records, and historical photos, Hammack and colleagues concluded that the aeromagnetic technique is 100% effective for

geolocation of old, abandoned wells (in their study site, those wells which were spudded or whose drilling was initiated before 1926) but only 82% effective for geolocation of more recently drilled wells (wells spudded in a second oil boom in the area after 1965), although lower flight altitudes would likely increase detection¹⁴⁻¹⁵. The authors speculated that the smaller casing diameter and different metallurgy in newer wells decreased the intensity of the magnetic anomaly associated with the steel-cased well.

Magnetic surveys for well localization have been conducted both aerially and on the ground. Aerial surveys are useful for surveying large swaths of land quickly, but sacrifice signal intensity and resolution¹³⁻¹⁴. In one of the first reports of using airborne magnetic surveys to identify abandoned wells, Frischknecht calculated that at a density of 770 wells/km² (2000 wells/mi²), the average spacing between wells would be near 36 m, requiring a flight pattern of 15 m line spacing, 15 m aboveground¹⁶⁻¹⁷. A complementary ground-based magnetic survey may be necessary in areas of high well density, but while well anomalies can be located to within 1-2 m in ground-based magnetic surveys¹³, such resolution comes at the expense of increased time and labor. Alternatively, new technologies have enabled the use of unmanned aerial vehicles (UAVs) to carry out aeromagnetic surveys. Dependent on the development of lighter magnetometers, UAV-based surveys mediate between loss of resolution in aerial surveys by permitting lower altitude flights without the time intensiveness of a ground-based survey¹⁵. The current industry outlook indicates economic success is possible with an oil field density of 3 wells/km², far below Frischknecht's calculated threshold for resolution of individual wells using a drone- or helicopter-based survey¹⁸. However, as historic oil fields were drilled without current geophysical knowledge, wells were often drilled at a much higher density and as such, these sites may require the increased resolution of a UAV survey.

Here, we review six manned helicopter aeromagnetic surveys conducted in Pennsylvania and Wyoming, looking specifically at the number of wells detected in the study areas and comparing those to state and proprietary databases. We make predictions based on these sites about the number of wells – including abandoned wells – in each state and nationally. These studies individually have the power to inform prospectors and researchers as to the location of active and abandoned wells, and together

103 demonstrate the power of aeromagnetic surveys to identify previously undocumented wells in diverse
104 landscapes.
105

METHODS

Site selection. The following six sites were incorporated into the analysis based on access to the location for aeromagnetic studies and previously collected aeromagnetic data: Hillman State Park (HSP) in southwestern Pennsylvania^{9, 19-20}, MCC Partners Oil Field (MCC) in southwestern Pennsylvania^{19, 21}, Ole Bull State Park (OBSP) in north-central Pennsylvania^{19, 21-23}, Oil Creek State Park (OCSP) in northwestern Pennsylvania¹⁹, Salt Creek Oil Field (SCOF) in eastern-central Wyoming²⁴⁻²⁵, and Teapot Dome Oil Field in eastern-central Wyoming (Figure 2)²⁵⁻²⁶.

Data collection and analysis. Aeromagnetic surveys were conducted as previously described between 2008 and 2016 and data were processed according to the previously documented studies^{9, 14-15, 19, 24, 26}. While some details of each data collection differed based on site-specific concerns (topography, availability of operator), the general procedure was as follows: A helicopter equipped with two Scintrex CS-3 Cesium vapor magnetic sensors (to remove geomagnetic drift) and high-resolution GPS (to provide georeferenced

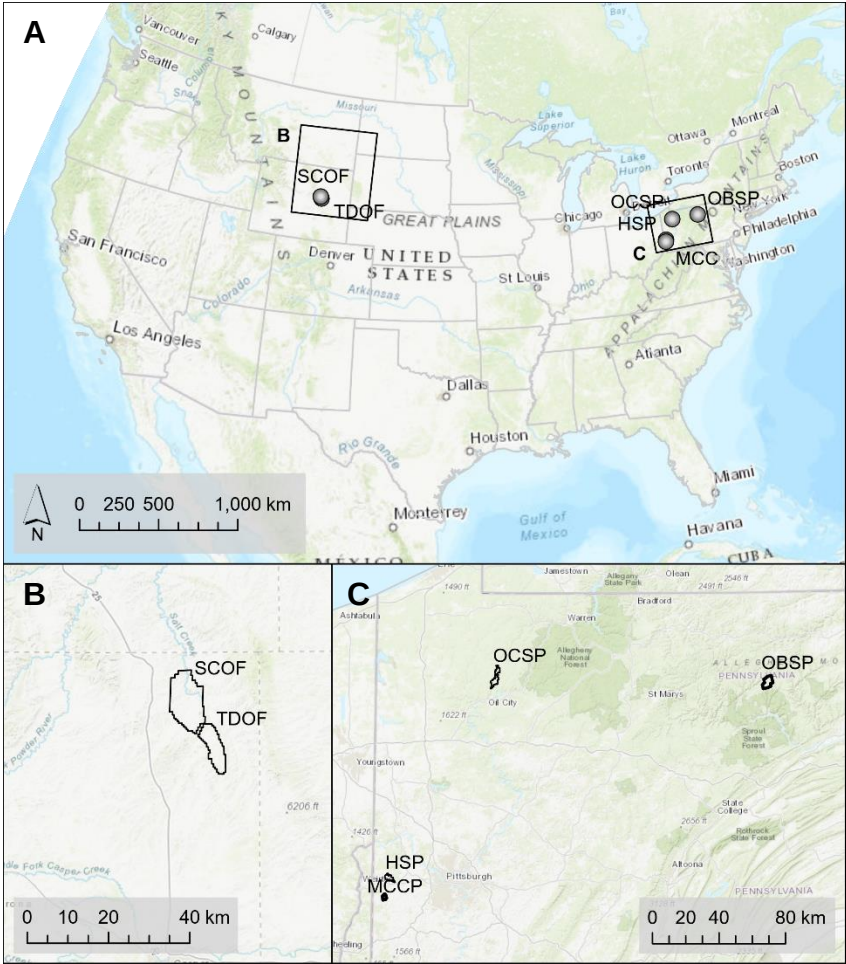


Figure 2. Locations of the aeromagnetic survey sites. (A) Locations in the Continental United States (CONUS). (B) Inset showing locations of survey sites in Wyoming. (C) Inset showing locations of survey sites in Pennsylvania. HSP, Hillman State Park; MCC: MCC Partners Oil Field; OBSP, Ole Bull State Park; OCSP: Oil Creek State Park; SCOF, Salt Creek Oil Field; TDOF, Teapot Dome Oil Field.

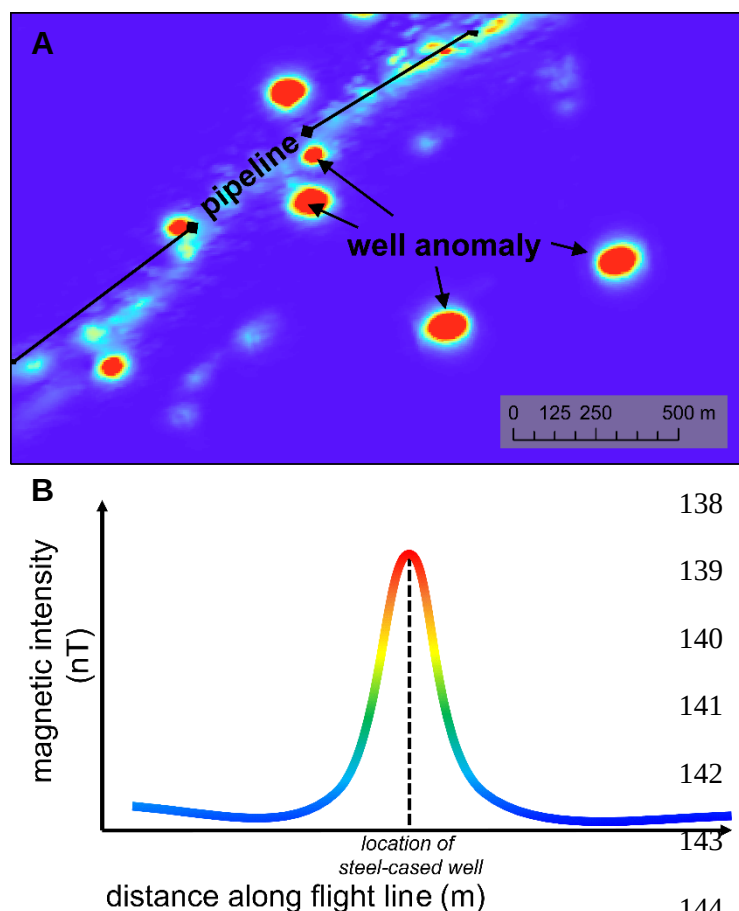


Figure 3. Magnetic anomalies associated with wells in heat maps. (A) Heat map from processed aeromagnetic data showing the presence of steel-cased wells (red bull's-eyes) and a pipeline (blue diagonal line). (B) Representative graph of distance and magnetic intensity, with the highest intensity centered above a steel-cased well.

magnetic data) flew parallel flight lines at an elevation of 30-50 m above ground. The height of the helicopter was determined using a laser altimeter. Specifications of collection of the data collection at each site are tabulated in Table S1. The data were preprocessed by the contractor to compensate for errors from the magnetic field of the helicopter itself, diurnal variation in the Earth's magnetic field, and sensor drift using a continuously recording magnetic base station. Standard processing using Geosoft software (Geosoft, Inc, Toronto, CA) yielded a heat (or "bump") map with a cell size of 20% of the line spacing of the flight pattern, in which the bull's-eye

anomalies were centered above steel-cased wells (Figure 3). Analytic signal (AS) processing of the data further enhanced well anomaly differentiation from the background^{9, 27}. A Blakely algorithm was used to extract point locations from the AS grid of magnetic data²⁸, which were manually cross-checked by comparing point locations to bull's-eye locations in the heat maps.

Geographic Information System (GIS) Analysis. Geosoft was used to process the magnetic data to resolve defined points for the magnetic anomaly. Auxiliary data included aerial photographs, historic maps, Light Detection and Ranging (LiDAR) elevation data, and cultural features such as roads, trails, and utilities. Auxiliary data were obtained from sources including the Drake Well Museum, Pennsylvania Spatial Data Access (PASDA)²⁹, Penn Pilot³⁰, the Pennsylvania Department of Conservation of Natural Resources, the

Pennsylvania Historical and Museum Commission, Wyoming State Archives, Wyoming State Historical Society, the Wyoming State Geological Survey, and the Wyoming Geospatial Hub. Geospatial analyses were performed using ArcMAP and ArcGIS Pro 2.4 (Esri, Redlands, CA, USA). Maps were designed in and exported from ArcGIS Pro 2.4.

Database well information. Data for wells located at each study site were downloaded from DrillingInfo (DI) (www.drillinginfo.com),³¹ the Pennsylvania Geological Survey's Exploration and Development Well Information Network (EDWIN) (www.dcnr.pa.gov)³², and the Wyoming State Geological Survey's Oil and Gas Map of Wyoming (www.wsgs.maps.arcgis.com)³³ in July 2019. Oil and gas wells of all statuses (i.e., active, inactive, abandoned) were included in the datasets. Wells of other production types (i.e., injection or storage) were not included as many are lined with plastic rather than ferromagnetic material. Wells in the databases that were spudded after the date of the magnetic survey were filtered from the results. Water well datasets were downloaded from the Pennsylvania Groundwater Information System (PaGWIS) (www.dcnr.pa.gov/Conservation/Water/Groundwater/PAGroundwaterInformationSystem/Pages/default.aspx)³⁴ and the Wyoming State Geological Survey's Groundwater Atlas of Wyoming (www.wsgs.maps.arcgis.com)³⁵ in September 2019.

% Difference. To determine the difference between the databases, a parameter defined as percent difference was calculated according to the standard formula: $x = \frac{|a-b|}{(a+b)/2}$, where x is the percent difference and a and b are the numbers of well locations in the databases being compared.

% Error. To determine the difference between the databases and the actual number of wells determined in the magnetic survey, a parameter defined as percent error was calculated according to the standard formula: $x = \left| \frac{b-a}{b} \right|$, where x is the percent error, a is the number of wells in the database, and b is the number of magnetic points.

Outlier Analysis. Outliers were determined by calculating the 1st quartile (1Q), 3rd quartile (3Q), and the interquartile range (IQR). Any value $> ([1.5 \times IQR] + 3Q)$ or $< (1Q - [1.5 \times IQR])$ was not included in further analysis and considered an outlier.

Nationwide and Statewide Well Estimates. After testing for and removing outliers, the nationwide well estimate was calculated by taking the mean % error and back-calculating for b in the % error equation using $a = 3,001,204$, or the theoretical number of wells in the continental United States (CONUS) in the DI database. State estimates were calculated similarly, but using both DI and State database counts. Standard deviation of the mean was calculated using the estimates for each site individually.

RESULTS AND DISCUSSION

Database comparison. We first compiled and compared publicly available datasets for abandoned and active wells at each of the six study sites. Wyoming and Pennsylvania both maintain records of well locations and statuses (Wyoming State Geological Survey and Pennsylvania Geological Survey)^{33, 36}. In all six locations, the state databases contained more wells than the national DrillingInfo (DI) database, which integrates well information from multiple sources (Table 1)³¹. The percent difference between databases across all sites was 66.6% $\pm$ 70.6%. The difference is most pronounced at HSP, where only one well is recorded in the DI database while PA has records for 33 wells. The two WY sites, SCOF and TDOF, have smaller percent differences than the locations in PA.

Differences between the datasets may reflect the limited information state agencies share with private dataset curators such as DI. Conversely, the DI datasets, which integrate wellhead information from multiple undisclosed sources, could be a more accurate reflection of the number of wells by eliminating duplicate or unreliable information. Abandoned well data has only been recently incorporated into DI³⁷, suggesting that continuing updates may bring the datasets into greater agreement.

Location	Size (km ²)	No. DrillingInfo Database Well Locations	DI Well Density (wells/km ²)	No. State Database Well Locations	State Well Density (wells/km ²)	% Difference
HSP	15.71	1	0.1	33	2.1	188%
MCC	0.57	6	10.5	9	15.8	40%
OBSP	1.56	10	6.4	54	34.6	138%
OCSP	28.97	955	33.0	1163	40.1	20%
SCOF	88.22	3703	42.0	4231	48.0	13%
TDOF	39.36	1356	34.5	1369	34.5	1%
Mean $\pm$ Standard Deviation	29.07 $\pm$ 29.87	1005 $\pm$ 1317	21 $\pm$ 16	1143 $\pm$ 1488	29 $\pm$ 16	66.6% $\pm$ 70.6%

Table 1. Study sites for magnetic surveys by area, well counts, and well densities according to state and DrillingInfo (DI) databases.

Since modern GPS technology has only been available since the 1970s³⁸, we initially hypothesized that sites with historic wells would have less accurate well location records than sites with more recent drilling activity. However, the average age of the wells at any of the six survey areas does not correlate to the percent difference between databases (Figure 4), meaning that sites with older wells do not have greater disparity. The analysis is limited due to the lack of documented well ages in many cases; for example, none of the wells at HSP have known completion or spud dates in either the state or DI database, nor do wells from the state database at MCC. OCSF, where many of the wells date from the mid-1800s but lack known completion dates, is particularly skewed to only newer wells. Thus, any slight degree of correlation between

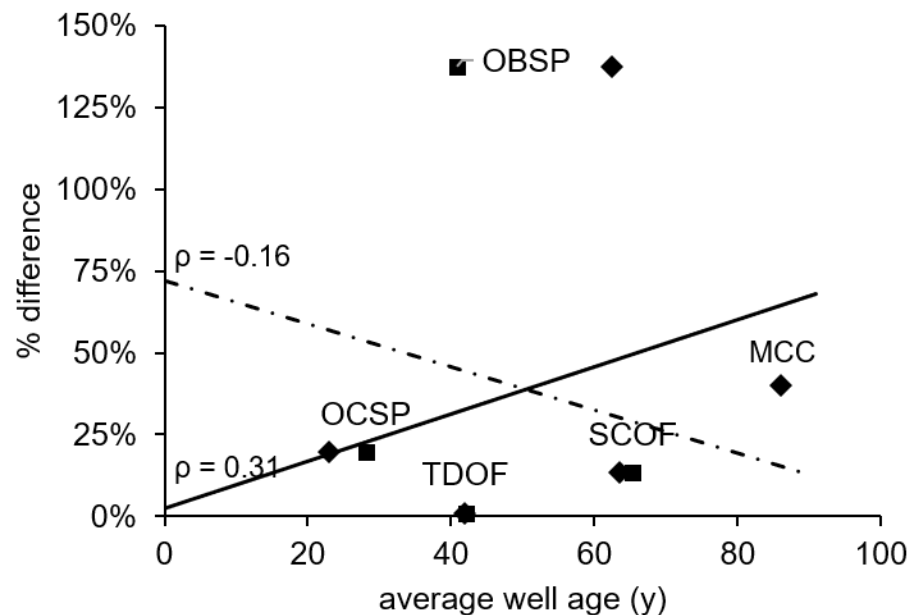


Figure 4. Average well age and % difference between databases. ♦, average age of wells in DI database. ■, average age of wells in state database. Best-fit lines with Pearson correlations (ρ) for DI database (solid line) and state database points (hatched line).

well age and percent difference between databases will be skewed because of the lack of known completion dates. Only at SCOF and TDOF did >90% of the wells have completion dates in both databases. From the limited availability of data, it is difficult to determine if well age influences reliability of well counts in a certain area. Additionally, sites having more complete records for wells do not tend to have smaller percent differences in well counts between databases.

216 *Consideration of water wells.* Although the Pennsylvania locations were remote, containing few to no
217 residential sites which often have associated water wells, we investigated the possibility of detecting water
218 wells in the aeromagnetic surveys in Pennsylvania and Wyoming. Similar to oil and gas wells, water wells
219 sometimes contain ferromagnetic casing, resulting in a false positive for the presence of an oil or gas well⁸.
220 Databases for water wells are, similar to oil and gas well databases, historically inaccurate³⁹. However, the
221 best available data from Pennsylvania's PaGWIS and Wyoming's Groundwater Atlas were used to
222 determine the number of water wells in each study site³⁴⁻³⁵. The coordinates were then overlaid on the
223 magnetic datasets to determine if the coordinates corresponded to a magnetic anomaly indicating a well.
224 Of the 104 wells located within the six study sites, 35 (34%) were located less than 50 m from a magnetic
225 point (Table 2), suggesting that either the coordinates for many water wells are very imprecise or that the
226 majority of water wells are lined with non-ferromagnetic materials and were invisible to the airborne
227 magnetometer. Only 0.5% of all magnetic points could be attributed as a water well across all locations
228 using a standard of <50 m error between magnetic coordinates and database coordinates (Table 2). Given
229 the small number of potential false positives due to water wells, we therefore elected to not factor in water
230 wells when considering well counts.

231 *Well density.* According to Frischnecht and colleagues, a density of 770 wells/km² can be spatially resolved
232 at a magnetometer height of 15 m¹⁶. Ingraffea and colleagues stated that shale gas wells are only profitable
233 at a density of 1 well per 80 acres, or 3 wells/km^{2,18}. This is well below the resolution limit of a magnetic
234 survey flown at 15 m aboveground, indicating that a higher flight pattern with reduced resolution would
235 still be useful in a region with both historic and more modern oil activity. Height of surveys is determined
236 not by density of wells in the region but by geographic limitations (i.e., tree height, topography) and aerial
237 limitations (i.e., minimum height for maintaining a safe and continuous survey).

Location	No. of Water Wells	No. <50 m from Magnetic Point	% Total Magnetic Points
HSP	1	0	0.00%
MCC	3	2	2.00%
OCSP	6	4	0.43%
OBSP	44	0	0.00%
SCOF	36	29	0.56%
TDOF	17	0	0.00%
Total	104	35	0.40%
Mean			0.50% $\pm$ 0.71%

Table 2. Number of water wells in each study site, including those within 50 m of a magnetic well location, which was used to calculate the percent of total magnetic points that may be attributable to water wells at each location.

In each location, the number of wells per km² was calculated to determine if the resolution of the aerially collected magnetic data would be sufficient to map the locations of potential wells; for both state and DI databases, the calculated well density falls well below the threshold (Table 1). The site with the lowest tabulated well density is HSP, with 0.1 wells/km² from the DI database and 2.1 wells/km² from the state database. The highest well density was at SCOF, with 42 and 48 wells/km² from the DI and state databases, respectively. The average well density using magnetic detection was 63 $\pm$ 53 wells/km², with the lowest density recorded at HSP (11 wells/km²) and highest at TDOF (59 wells/km²) (Table 3). Although the density across each location is below the threshold required for individual well resolution in an aeromagnetic survey, well density is not homogeneous at each site (Supplementary Figures S1 and S2). For example, regions of increased data density occur near Oil Creek in OCSP (Figure 5), creating overlapping datapoints that are difficult to resolve. Dense clusters of wells are common in historic oil fields such as OCSP, where prospectors had little understanding of geology and drilled wells much closer together than in modern oil fields. The overlapping regions of high magnetic intensity make automatic peak picking challenging, resulting

in errors such as over- or under-estimation of well counts and incorrect geolocation. (A subsequent ground magnetic survey in one area of OCSF resolved an additional 20 wells, but the data are not included in this analysis.) Despite small regions with extremely high well density such as the site

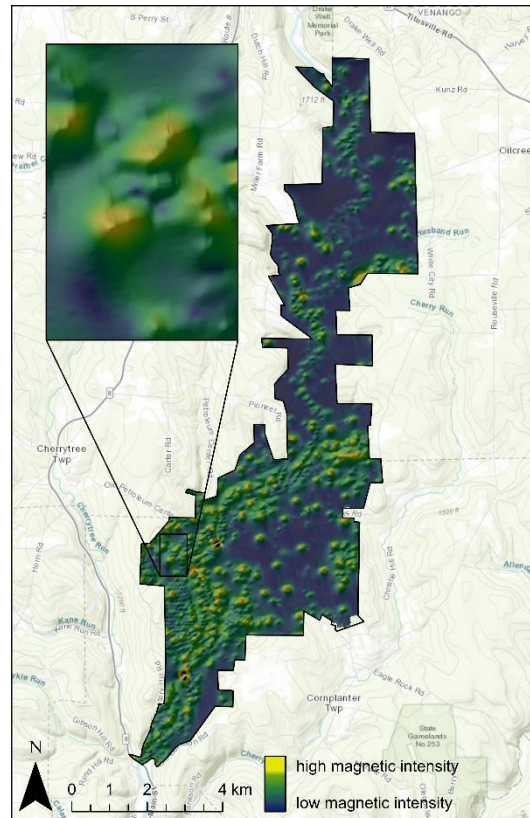


Figure 5. Magnetic heat map of OCSF and inset showing region of high well density and low resolving power into individual wellheads.

shown in Figure 5, individual wells could be resolved across most of the surveyed area at each study site.

Sensitivity of detection of wells is also quite high. In particular, wells buried under at least 30 m of mine overburden were still detectable at HSP⁹. At HSP, strip mining for coal dramatically changed the topography, and many wells were buried under the massive overburden removed during mining between 1914 and 1968⁴⁰. Although conservation and restoration efforts began as early as 1937, contouring and replanting the area resulted in a dramatically different topology than was present when the oil wells were

262 in operation (Supplementary Figure S3). Nonetheless, the aeromagnetic survey was able to detect buried
 263 well casings, even those that are not visible at the surface or buried deep underground. During field

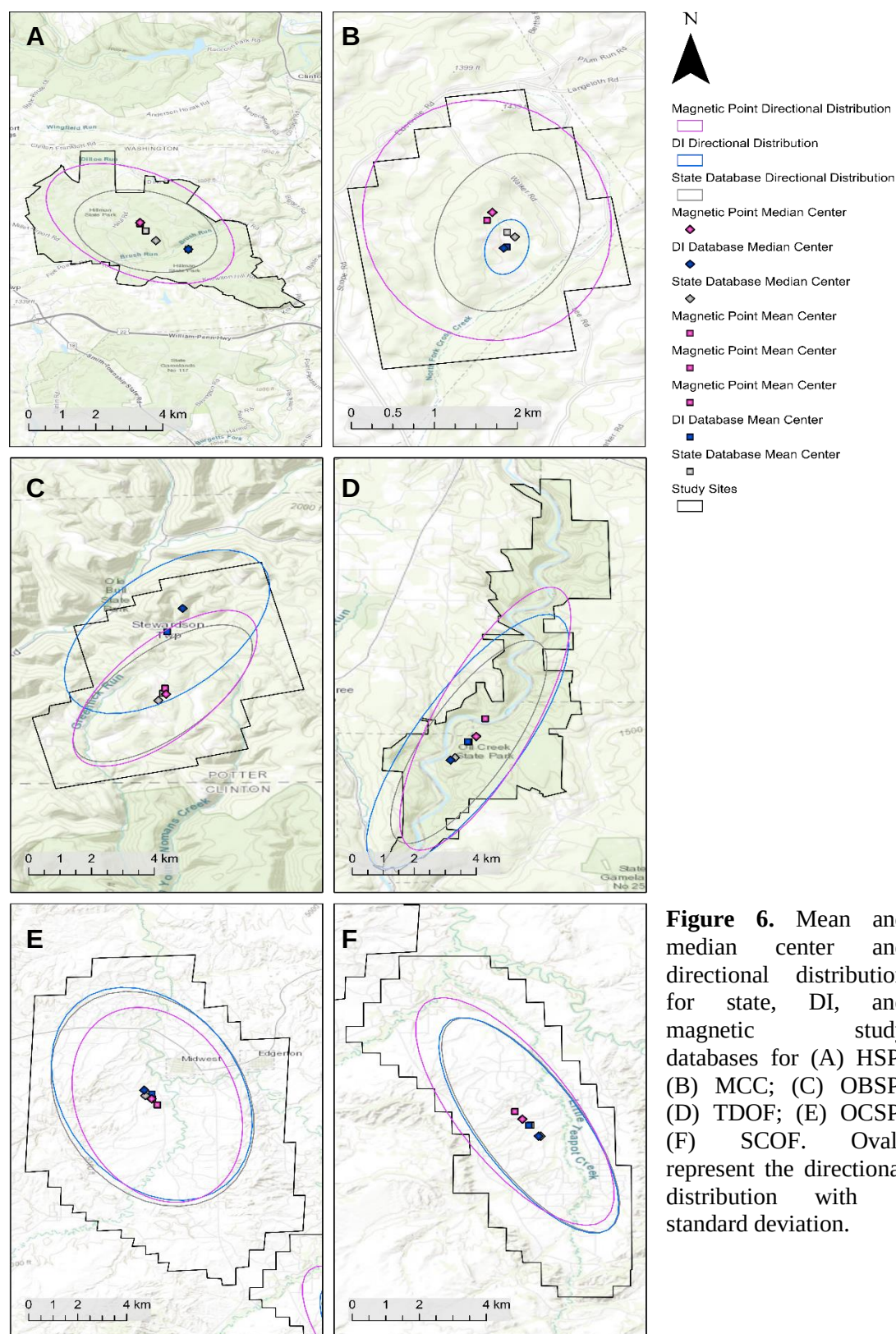


Figure 6. Mean and median center and directional distribution for state, DI, and magnetic study databases for (A) HSP; (B) MCC; (C) OBSP; (D) TDOF; (E) OCSP; (F) SCOF. Ovals represent the directional distribution with 1 standard deviation.

verification of the aeromagnetic survey at HSP, some sites with no visible surface feature were excavated to the well casing. While time and labor intensive, some sites that were shallowly excavated resulted in the discovery of a buried wellhead⁹.

Spatial distribution of wells. Because of the high number of magnetic well sites at some of the locations (Table 3), maps showing locations of wells are difficult to interpret visually. However, based on the magnetic data it is clear that wells are often clustered rather than equally distributed across the landscape (Supplementary Figures S1 and S2). In all but OBSP, the mean and median distributions are similar for datasets and the magnetic study; however, the DI database skewed significantly from the state and aeromagnetic datasets due to a lack of datapoints in the southeastern portion of the study site. At HSP, MCC, and TDOF, the magnetic points are more widely distributed than the databases would suggest (Figure 6). Modern wells are often drilled with 100 m separation in a grid-like layout, such as at SCOF and TDOF (Supplementary Figures S1 and S2). Interestingly, at SCOF, wells appear to be drilled more densely in certain Public Land Survey System (PLSS) sections, evenly interspersed with less heavily drilled PLSS blocks.

Estimation of number of wells statewide and nationwide. Many researchers have speculated on the number of wells drilled in the state of Pennsylvania, with estimates from 300,000 up to 500,000^{6, 41-44}. Using a combination of historical records, current databases, and farm line maps, Dilmore and colleagues estimated a total of 330,560 completions in Pennsylvania from 1859 through 2015⁶. The Pennsylvania EDWIN database has 183,995 documented wells, in close agreement with the DI count of 156,088 but well below published estimates. We do not find it surprising, therefore, that state and DI databases would underrepresent the number of wells found at many of the study sites. Except at SCOF, the magnetic surveys located more wells than either database, although it should be noted that magnetic intensity thresholds influence the number of wells located in an aeromagnetic survey (Table 3)^{24, 26}. The average percent error between magnetic and state counts was 45.6% $\pm$ 29.5%, while the average percent error between magnetic and DI counts was 58.5% $\pm$ 36.4%. While the values of the average percent error values are not statistically significant, only at OCSP was the percent difference smaller for the DI count compared to the magnetic

counts. In general, therefore, while both datasets underestimated the number of wells in a study site compared to the aeromagnetic survey results, the state database counts were closer to the actual number of wells found in the magnetic surveys.

OCSP presents a unique case in wellfinding efforts that may not directly apply to other sites. Covering part of the world's first commercial oil field, OCSP is home to some of the oldest oil wells ever drilled⁴⁵. Some of the earliest oil wells, which were often shallow (<100 m deep), were shored at the surface with wood conduit casing rather than steel⁴⁶. These wells, lacking near-surface ferromagnetic material would not be detected in a magnetic survey. Photographic evidence suggests that as many as 17 wells were drilled on a hillside within what is now OCSP, yet magnetic contours of the same hillside indicate only two remaining magnetized well casings (Supplementary Figure S4).

Location	Size (km ²)	No. Magnetic Well Locations	Magnetic Well Density (wells/km ²)	No. State Database Well Locations	% Error Magnetic & State Database	No. DI Database Well Locations	% Error Magnetic & DI Database
HSP	15.71	165	11	33	80.0%	1	99.4%
MCC	0.57	100	175	9	91.0%	6	94.0%
OBSP	1.56	66	42	54	18.2%	10	84.8%
OCSP	28.97	931	32	1163	24.9%	955	2.6%
SCOF	88.22	5154	58	4231	17.9%	3703	28.2%
TDOF	39.36	2333	59	1369	41.3%	1356	41.9%
Mean	29.07	1458	63	1143	45.6%	1005	58.5%
±Standard Deviation	±29.87	±1833	±53	±1489	±29.5%	±1317	±36.4%

Table 3. Comparison of number of well locations calculated from aeromagnetic surveys compared to state and DrillingInfo (DI) databases.

Some recent publications have estimated the number of abandoned wells in the CONUS to be between 2-3 million⁴⁷⁻⁴⁸. Given that the total number of O&G wells of all status types (active, inactive, abandoned, plugged) in the DI national dataset is 3,001,204 wells, of which only 19.2% are classified as abandoned or inactive³¹, there are large discrepancies between database well counts and estimates based on number of well sites from aeromagnetic surveys.

Using the percent error between number of magnetic sites and database well locations, we estimated the number of wells that could be present in Pennsylvania, Wyoming, and the continental United States (CONUS) (Table 4). Because all locations except OCSP had more magnetic sites than documented wells, our estimations likewise suggest more wells than in the current databases. The observations made at HSP using the DI dataset make it a clear outlier in the dataset and dramatically upwardly skew estimations; thus, we have elected to ignore the outlier (with a calculated 495 million wells) in estimations using the DI dataset.

In Pennsylvania, there are an estimated 466,000 wells based on the state database error, and 395,000 wells based on the mean error from the DI database (Table 4). These estimates are in good agreement with previously published estimates. The state and DI databases label 7.2% and 4.7% wells as abandoned, respectively, giving an estimated 34,000 and 19,000 abandoned wells in the state. In Wyoming, the calculated estimates were 181,000 – 182,000 with 23,000 – 25,000 abandoned wells.

The average number of calculated wells in the CONUS is 6.04 ± 19.97 million wells with 1.16 ± 3.84 million of those designated as abandoned wells. While the error associated with the estimation places the number of abandoned wells in agreement with other estimations, the lower estimation is likely the result of basing calculations on a dataset that is known to be lacking in abandoned well counts³⁷. The percentage of abandoned wells may be higher or lower than 19.2%, considering that 47.4% of wells in the Wyoming State Geological Survey database are abandoned, shut-in, or temporarily abandoned and 7.4% of the drilled wells in the Pennsylvania EDWIN database are abandoned, orphaned, or inactive^{33, 36}. While there are limitations to this analysis because of limited records for abandoned legacy wells, the percentage of wells that are abandoned or inactive in a region covered by an aeromagnetic survey could be determined using aerial

photographs or a field survey. Additionally, the overall analysis is limited to two specific regions which are not necessarily representative of O&G fields across the CONUS. Other predictions have relied solely on database mining to estimate the number of abandoned wells, while this study connects field-based evidence for abandoned wells to database counts. States with more recent drilling activity may have better records, some areas are more uniformly drilled than regions with more variable topography, and changes in well casing materials may influence the ability to detect wells using magnetometers. Estimates of well numbers

Location	PA*	WY	CONUS*
Mean % Error with State Database	44.70%	29.61%	
No. Database Well Locations (State)	183,995	117,443	
% Abandoned Wells**	7.2%	12.5%	
Estimated No. Wells	465,521	180,724	
Estimated No. Abandoned Wells	33,609	22,610	
Mean % Error with DI Database	60.48%	35.02%	50.29%
No. Database Well Locations (DI)	156,088	117,964	3,001,204
% Abandoned Wells**	4.7%	13.7%	19.2%
Estimated No. Wells	394,914	181,525	6,037,587
Estimated No. Abandoned Wells	18,707	24,938	1,159,689

*excludes HSP data for DI dataset

**does not include plugged and abandoned wells

Table 4. Estimated number of wells in Pennsylvania, Wyoming, and the Continental United States (CONUS) based on percent error calculations between databases and magnetic points at study sites.

– both active and abandoned – are useful for broadly predicting well counts in a region but uncertainty is high and therefore the estimations are not recommended to be used to predict well density.

State and proprietary databases for active and abandoned wells record the locations of active, inactive, and abandoned oil and gas wells. However, due to lack of regulation during the early years of the oil industry, many wells remain undocumented. Aeromagnetic surveys can detect the ferromagnetic casing of wells, even legacy wells that have been cut off at the surface and buried. The survey data can be processed

to resolve well-type magnetic features and produce coordinates for steel-cased wells based on magnetic intensity. This technique was used at six locations in Pennsylvania and Wyoming and compared to the state and propriety databases covering the regions. In most cases, more wells were detected than in either database, with the state databases more closely reflecting the wells detected from the surveys. While the number of magnetic anomalies can be influenced by magnetic intensity thresholds, which are specific to each location and its magnetic field, the strength of the monopole of ferromagnetic well casings makes near- and above-surface well features strikingly different from the surrounding magnetic field. The aeromagnetic surveys found more wells than recorded in state databases in five of the six locations. The EPA estimates of the number of abandoned wells in the United States to be 2-3 million, and the estimates based on the analysis of the Wyoming and Pennsylvania surveys are in good agreement with this estimation. It should be noted that there are significant uncertainties in this calculation, and it is likely that the number of abandoned wells in the CONUS is underestimated.

Limitations to determining the location of abandoned wells with aeromagnetic surveys include (1) the cost of airborne surveys; (2) spatial resolution issues in the presence of other ferromagnetic infrastructure⁴⁹, (3) detection of both active and abandoned wells; and (4) lack of detection of wells whose casings have been removed or were constructed using wooden conduit casing^{7, 50}. Water wells, whose documentation across the nation is limited^{4, 39}, may also have ferromagnetic casing and could skew the data slightly especially in regions with higher population density⁴⁹. Yet airborne analyses provide coverage over a large region with enough resolution to identify individual well locations with reasonable accuracy to locate the site in the field. Aeromagnetic surveys are useful in mapping the locations of both active and inactive wells in open, relatively flat regions such as western Wyoming and in regions with dense foliage and difficult terrain, such as western Pennsylvania. As drone surveys become cheaper and battery lives extend survey ranges, aeromagnetic surveys may become an important tool for well exploration. Magnetic surveys for wellfinding thus present an important advancement in the discovery of abandoned O&G wells for the purpose of plugging and reclamation, thereby preventing potential groundwater and atmospheric contamination through loss of well integrity^{10, 51-52}.

SUPPORTING INFORMATION

The Supporting Information is comprised of four figures and one table.

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391 **ABBREVIATIONS**

392

393 CONUS: Continental United States

394 DI: DrillingInfo

395 EDWIN: Exploration and Development Well Information Network

396 EPA: Environmental Protection Agency

397 GIS: Geographic Information System

398 HSP: Hillman State Park

399 LiDAR: Light Detection and Ranging

400 NG: Natural gas

401 OCSP: Oil Creek State Park

402 OBSP: Ole Bull State Park

403 O&G: Oil and natural gas

404 PaGWIS: Pennsylvania Groundwater Information System

405 PASDA: Pennsylvania Spatial Data Access

406 PLSS: Public Land Survey System

407 SCOF: Salt Creek Oil Field

408 TDOF: Teapot Dome Oil Field

409 UAV: Unmanned aerial vehicle

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REFERENCES

1. Wells, B. A. First Oil Discoveries. www.aoghs.org/petroleum-discoveries/.
2. U.S. Energy Information Administration, U.S. energy consumption by energy source, 2017. April 2018 ed.; U.S. Energy Information Administration: Washington, DC, USA, 2018; p Table 1.3.
3. Dusseault, M.; Jackson, R., Seepage pathway assessment for natural gas to shallow groundwater during well stimulation, in production, and after abandonment. *Environmental Geosciences* **2014**, *21* (3), 107-126.
4. Jasechko, S.; Perrone, D., Hydraulic fracturing near domestic groundwater wells. *Proc Natl Acad Sci USA* **2017**, *114* (50), 13138-13143.
5. Pennsylvania Department of Conservation and Natural Resources, Pennsylvania's Oil and Gas Act (Act 223). In *58 P.S. § 601.101*, updated 10/08/2016 ed.; Commonwealth of Pennsylvania: Harrisburg, PA, USA, 1984; Vol. § 78a.15. Application requirements.
6. Dilmore, R. M.; Sams, J. I., 3rd; Glosser, D.; Carter, K. M.; Bain, D. J., Spatial and temporal characteristics of historical oil and gas wells in Pennsylvania: Implications for new shale gas resources. *Environ Sci Technol* **2015**, *49* (20), 12015-23.
7. Taku Ide, S.; Friedmann, J.; J. Herzog, H. In *CO₂ leakage through existing wells: Current technology and regulations*, 8th Greenhouse Gas Technology Conference, Trondjheim, Norway, Trondjheim, Norway, 2006.
8. Frischknecht, F. C.; Grette, R.; Raab, P. V.; Meredith, J., Location of abandoned wells by magnetic surveys; acquisition and interpretation of aeromagnetic data for five test areas. U.S. Geological Survey: Washington, DC, USA, 1985.
9. Sams, J. I., 3rd; Veloski, G.; Diehl, J. R.; Hammack, R. W., Methods and challenges to locating legacy wells in western Pennsylvania: Case study at Hillman State Park. *Environ Geosci* **2017**, *24* (2), 95-112.
10. King, G. E.; Valencia, R. L., Environmental risk and well integrity of plugged and abandoned wells. In *SPE Annual Technical Conference and Exhibition*, Society of Petroleum Engineers: Amsterdam, The Netherlands, 2014; 17.
11. Oil and Gas Wells, Drilling Regulating Act of May 17, 1921. In *P.L. 912*, Legislature, P. S., Ed. Philadelphia, PA, USA, 1921; Vol. No. 322.
12. Butler, K. E.; Grace, M.; Boulay, D.; Black, A. In *Locating abandoned wells using ground and UAV magnetometry: examples from Stoney Creek Oil and Gas Field, New Brunswick, Canada*, SAGEEP 2019 - 32nd Annual Symposium on the Application of Geophysics to Engineering and Environmental Problems. Environmental & Engineering Geophysical Society: Portland, OR, USA, 2019; 425.
13. Frischknecht, F. C.; Raab, P. V., Location of abandoned wells with geophysical methods. US Geological Survey Geological Division. US Environmental Protection Agency: Las Vegas, NV, USA, 1983.
14. Hammack, R. W.; Veloski, G. A.; Hodges, D. G.; White, C. M., Methods for finding legacy wells in large areas. Office of Science, Department of Energy. National Energy Technology Laboratory: Pittsburgh, PA, USA, 2016; 28.
15. Hammack, R.; Veloski, G.; Sams, J., Using Drone Magnetic and LiDAR Surveys to Locate Unmarked, Abandoned Wells Prior to Unconventional Oil and Gas Development. In *Unconventional Resources Technology Conference, Houston, Texas, 23-25 July 2018*, Society of Exploration Geophysicists, American Association of Petroleum Geologists, Society of Petroleum Engineers: 2018; 188-196.
16. Frischknecht, F. C.; Muth, L.; Grette, R.; Buckley, T.; Kornegay, B., Geophysical methods for locating abandoned wells. In *Open-File Report*; US Geological Survey, Denver, CO, USA, 1983.
17. Phillips, J. D. In *Tools and Techniques for the Application of High-Resolution Aeromagnetic Surveys to Engineering Problems*, Symposium on the Application of Geophysics to Engineering and Environmental Problems 2001, Society of Exploration Geophysicists: Denver, CO, USA, 2001; AIR4-AIR4.

18. Ingraffea, A. R.; Wells, M. T.; Santoro, R. L.; Shonkoff, S. B. C., Assessment and risk analysis of casing and cement impairment in oil and gas wells in Pennsylvania, 2000–2012. *Proc Natl Acad Sci USA* **2014**, *111* (30), 10955.
19. Hammack, R.; Veloski, G.; Sams, J., Rapid methods for locating existing well penetrations in unconventional well development areas of Pennsylvania. In *Unconventional Resources Technology Conference, San Antonio, Texas, 20-22 July 2015*, Society of Exploration Geophysicists, American Association of Petroleum Geologists, Society of Petroleum Engineers: 2015; 1309-1324.
20. Pekney, N. J.; Diehl, J. R.; Ruehl, D.; Sams, J.; Veloski, G.; Patel, A.; Schmidt, C.; Card, T., Measurement of methane emissions from abandoned oil and gas wells in Hillman State Park, Pennsylvania. *Carbon Manag* **2018**, *9* (2), 165-175.
21. Hammack, R. W.; Veloski, G. A. *Methods for Finding Legacy Wells in Residential and Commercial Areas*. Office of Science, Department of Energy, National Energy Technology Laboratory: Pittsburgh, PA, USA, 2016.
22. Hammack, R. W.; Sams III, J. I.; Veloski, G. A.; Diehl, J. R. *Evaluation of a Rapid Airborne Method for Locating Legacy Wells in Pennsylvania*; Office of Science, Department of Energy. National Energy Technology Laboratory: Pittsburgh, PA, USA, 2016.
23. Hammack, R.; Veloski, G.; Hodges, G.; McLaren, S.; White, C. In *An evaluation of helicopter and ground methods for locating existing wells*, 19th EEGS Symposium on the Application of Geophysics to Engineering and Environmental Problems, 2006.
24. Kaminski, V.; Hammack, R. W.; Harbert, W.; Veloski, G. A.; Sams, J.; Greg Hodges, D., Geophysical helicopter-based magnetic methods for locating wells. *Geophysics* **2018**, *83* (5), B269-B279.
25. Veloski, G.; Hammack, R.; Lynn, R. In *Practical methods for locating abandoned wells in populated areas*, Symposium on the Application of Geophysics to Engineering and Environmental Problems 2007, Society of Exploration Geophysicists: 2007; 636-652.
26. Veloski, G. A.; Hammack, R. W.; Stamp, V. W.; Hall, R.; Colina, K., Helicopter Magnetic Survey at the Teapot Dome Oilfield (Naval Petroleum Reserve No. 3): A Case History. In *Symposium on the Application of Geophysics to Engineering and Environmental Problems 2008*, Environment and Engineering Geophysical Society: Denver, CO, USA, 2008; 13-25.
27. MacLeod, I. N.; Jones, K.; Dai, T. F., 3-D analytic signal in the interpretation of total magnetic field data at low magnetic latitudes. *Exploration Geophysics* **1993**, *24* (4), 679-688.
28. Blakely, R. J.; Simpson, R. W., Approximating edges of source bodies from magnetic or gravity anomalies. *Geophysics* **1986**, *51* (7), 1494-1498.
29. Pennsylvania Department of Conservation and Natural Resources, Pennsylvania LiDAR elevation program. Bureau of Topographic and Geologic Survey: 2013.
30. Pennsylvania Geological Survey, Penn Pilot Aerial Photograph Collection. United States Department of Agriculture: Agricultural Adjustment Administration Northeast Division. Penn Pilot: Harrisburg, PA, USA, 1939.
31. DrillingInfo, Wells Data. DrillingInfo, Ed. Austin, TX, USA, 2019.
32. Pennsylvania Department of Conservation and Natural Resources, Exploration and Development Well Information Network. Pennsylvania Geological Survey. Pittsburgh, PA, USA, 2019.
33. Toner, R.; Lynds, R.; Stafford, J.; Kehoe, K.; Pisel, J., Interactive Oil and Gas Map of Wyoming. Survey, Wyoming Geological Survey. 2019.
34. Pennsylvania Department of Conservation and Natural Resources, PaGWIS. Pennsylvania Geological Survey. 2019.
35. Stafford, J.; Taboga, K.; Toner, R., Groundwater Atlas of Wyoming. Survey, Wyoming Geological Survey. 2017.
36. Pennsylvania Geological Survey, Exploration and Development Well Information Network (EDWIN). Pennsylvania Department of Conservation and Natural Resources. Pittsburgh, PA, USA, 2019.
37. Rimmel, M., Personal Communication (Phone Conversation). DrillingInfo: Littleton, CO, USA, 2019.

38. Mai, T. Global Positioning System History.
https://www.nasa.gov/directorates/heo/scan/communications/policy/GPS_History.html (accessed May 15, 2019).
39. Perrone, D.; Jasechko, S., Dry groundwater wells in the western United States. *Environ Res Lett* **2017**, *12* (10), 104002.
40. Kerlin, N. History of Hillman State Park.
<https://www.dcnr.pa.gov/StateParks/FindAPark/HillmanStatePark/Pages/History.aspx> (accessed 4 August 2019).
41. Detrow, S., Perilous Pathways: How Drilling Near An Abandoned Well Produced a Methane Geyser. State Impact Pennsylvania. National Public Radio: 2012.
42. Kang, M.; Kanno, C. M.; Reid, M. C.; Zhang, X.; Mauzerall, D. L.; Celia, M. A.; Chen, Y.; Onstott, T. C., Direct measurements of methane emissions from abandoned oil and gas wells in Pennsylvania. *Proc Natl Acad Sci USA* **2014**, *111* (51), 18173-18177.
43. Pennsylvania Department of Environmental Protection, Pennsylvania's Plan For Addressing Problem Abandoned Wells and Orphaned Wells. Department of Environmental Protection, Bureau of Oil and Gas Management. Harrisburg, PA, USA, 2000.
44. Mitchell, A. L.; Casman, E. A., Economic Incentives and Regulatory Framework for Shale Gas Well Site Reclamation in Pennsylvania. *Environ Sci Technol* **2011**, *45* (22), 9506-9514.
45. Saint-Vincent, P. M. B.; Sams III, J. I.; Reeder, M.; Hammack, R. J.; Veloski, G. A.; Mundia-Howe, M.; Pekney, N. J., Rediscovery of Abandoned Wells in the World's First Oil Field. *Esri Story Maps* **2019**.
46. Henry, J. D., *History and romance of the petroleum industry, Chapter 4: Discovery of oil in brine wells and its use in medicine*. Bradbury, Agnew: London, 1914; 392.
47. Brandt, A. R.; Heath, G. A.; Kort, E. A.; O'Sullivan, F.; Pétron, G.; Jordaan, S. M.; Tans, P.; Wilcox, J.; Gopstein, A. M.; Arent, D.; Wofsy, S.; Brown, N. J.; Bradley, R.; Stucky, G. D.; Eardley, D.; Harriss, R., Methane leaks from North American natural gas systems. *Science* **2014**, *343* (6172), 733-735.
48. Townsend-Small, A.; Ferrara, T. W.; Lyon, D. R.; Fries, A. E.; Lamb, B. K., Emissions of coalbed and natural gas methane from abandoned oil and gas wells in the United States. *Geophys Res Lett* **2016**, *43* (5), 2283-2290.
49. Jachens, R. C.; Webring, M. W.; Frischknecht, F. C. *Abandoned-well study in the Santa Clara Valley, California*; 86-350; 1986.
50. Bowman, I. *Well-drilling methods, Water-Supply Paper 257*; US Geological Survey, Department of the Interior. Government Printing Office, Washington, DC, USA, 1911.
51. Davies, R. J.; Almond, S.; Ward, R. S.; Jackson, R. B.; Adams, C.; Worrall, F.; Herringshaw, L. G.; Gluyas, J. G.; Whitehead, M. A., Oil and gas wells and their integrity: Implications for shale and unconventional resource exploitation. *Mar Petrol Geol* **2014**, *56*, 239-254.
52. Beattie, S.; Byron, L. A.; Catalano, A. D.; Pelepko, S.; Swantk, R., Pennsylvania legacy well integrity and emissions study. In *Geological Society of America 52nd Northeastern Annual Section/51st North-Central Annual Section Joint Meeting*, Pittsburgh, PA, 2017; Vol. 49, 47-3.
53. West Virginia Code. Chapter 22: Environmental Resources. Article 6. Office of Oil and Gas. §22-6-24. Methods of plugging well. West Virginia Legislature: Charleston, WV, USA.
54. Ohio Administrative Code. Chapter 1501:9-11 Plugging of Wells. Division of Mineral Resources Management – Oil and Gas. Columbus, OH, USA.
55. Oil & Gas Regulations and Statutes. 805 KAR Chapter 1:060. Plugging wells. Kentucky Energy and Environment Cabinet. Commonwealth of Kentucky, Lexington, KY, USA.
56. Code of State Regulations. Rules of Department of Natural Resources. Title 10: Department of Natural Resources. Division 50. Oil and Gas Council. Missouri Secretary of State, Jefferson City, MO, USA.

57. Code of Colorado Regulations. Title 34. Mineral Resources. Article 60. Oil and Gas Conservation. § 34-60-100 Oil and Gas Conservation Act of the State of Colorado. Colorado Secretary of State, Denver, CO, USA.
58. The Laws of New York. Environmental Conservation. Article 23: Mineral Resources. The New York State Senate, Albany, NY, USA.
59. California Code of Regulations. 14 CA ADC § 1723.1. Plugging of Oil or Gas Zones. California Secretary of State, Sacramento, CA, USA.
60. Oklahoma Administrative Code. Title 165: Corporation Commission. Chapter 10: Oil and Gas Conservation. Oklahoma Secretary of State, Oklahoma City, OK, USA.
61. Kansas Administrative Regulations. K.A.R. 82-3-111. General Rules and Regulations for the Conservation of Crude Oil and Natural Gas. Kansas State Legislature, Topeka, KS, USA.
62. Texas Administrative Code. Title 16: Economic Regulation. Part 1: Railroad Commission of Texas, Chapter 3: Oil and Gas Division. §3.14 Plugging. Office of the Secretary of State, Austin, TX, USA.
63. Louisiana Administrative Code. Title 43: Natural Resources. Part XIX. Office of Conservation, Chapter 1: General Operations. §137. Plugging and Abandonment. Office of Conservation. Office of the State Register, Baton Rouge, LA, USA.
64. Alaska Oil and Gas Laws and Regulations. Title 20 Miscellaneous Boards and Commissions. Chapter 25. Alaska Oil and Gas Conservation Commission. AAC 25.112. Well plugging requirements. The Alaska State Legislature, Juneau, AK, USA.
65. Illinois Administrative Code. Title 62: Mining. Chapter 1. Department of Natural Resources. §240. The Illinois Oil and Gas Act. Illinois General Assembly, Springfield, IL, USA.
66. Nebraska State Codes. Title 267. Nebraska Oil and Gas Conservation Commission. Chapter 3. Drilling, Development, Producing and Abandonment. Nebraska Legislature, Omaha, NE, USA.
67. Code of Alabama. Title 9. Conservation and Natural Resources. Chapter 17. Oil and Gas. §1 Conservation and Regulation of Production. Alabama Legislature, Montgomery, AL, USA.
68. Idaho Statutes. Title 47. Mines and Mining. Chapter 3. Oil and Gas Wells – Geologic Information, and Prevention of Waste. Idaho Legislature, Boise, ID, USA.